

Swoosh

A Production-Hardened URL Shortener

Live: url-shortener-api-jcbx.onrender.com

Built with FastAPI + Neon PostgreSQL

Hardened with validation, rate limiting, security headers, 26 tests

Developed using spec-driven workflow and Addy Osmani's agent-skills

Why Swoosh?

Problem: Long URLs are ugly, break in chats, and hide analytics.

Solution: Short, custom, trackable links with a clean glassmorphism UI.

What makes it production-ready:

- Password-protected access (show/hide toggle + logout)
- Input validation (Pydantic)
- Rate limiting (30 req/min per IP)
- Security headers on every response
- Structured JSON errors
- URL deduplication
- Click tracking + link management

How I Built It

Workflow: Spec → Plan → Build → Review → Ship (agent-skills lifecycle)

Step	Tool	What happened
Spec	Claude Code	Wrote structured requirements doc
Plan	Claude Code	Broke into 6 steps with checklist
Build	Claude Code	Incremental implementation, test-driven
MCP	mcp-server-sqlite	Live DB inspection during dev
Skill	url-api-contract	Enforced API consistency
Agent	url-tester	Autonomously verified endpoints

Architecture

Browser → FastAPI (Render) → Neon PostgreSQL

Rate Limit
Validation
Security

30 req/min per IP
Pydantic validators
Anti-XSS, anti-clickjacking

Endpoints:

- `POST /api/shorten` — create short URL (dedup + rate limit)
- `GET /<code>` — 302 redirect + click tracking
- `GET /api/links` — list all saved links
- `DELETE /api/links/<code>` — remove a link
- `GET /api/health` — monitoring

What Got Hardened

Area	Before	After
URL validation	<code>startswith("http")</code>	Pydantic HttpUrl, 2048 char limit
Custom codes	No limits	3-20 chars, alphanumeric, reserved words blocked
Rate limiting	None	30/min on POST via slowapi
Password	None	Password required for POST/DELETE
Errors	Raw HTTPException	Structured JSON everywhere
Database	SQLite, open/close per request	Context managers, Neon PostgreSQL
Config	Hardcoded	env. driven

What I'd Do Next

✅ Deployed — live on Render + Neon PostgreSQL (free, permanent)

Next steps:

- Add authentication for personal link management
- Add geographic analytics (clicks by country)
- Custom domain with HTTPS

GitHub: github.com/ahk1542001-wq/url-shortener-api

Thank you! 🌟